Curriculum Vitae

Siddharth Setlur

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Personal Information

Name:	Siddharth Setlur	Phone:	+44 735 609 38 90
Date of Birth:	18. Dec 2000	Languages:	English (first), German (fluent)
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Education

Sept 2024 – **PhD at CDT in Algebra, Geometry and Quantum Fields, University of Edinburgh**
Supervised by Dr. Darrick Lee & Dr. Tudor Dimofte. Fully funded by the UKRI ESPRC.

Sept 2022 – August 2024 **Master of Science (with distinction) at ETH Zürich in Mathematics**
MSc Thesis: *Optimizing Functions Based on Signed Barcodes*
Supervised by Dr. Sara Kališnik & Dr. Steve Oudot.

Sept 2019 – Aug 2022 **Bachelor of Science at ETH Zürich in Mathematics**
BSc Thesis: *Different Theoretical Perspectives on Persistent Homology*
Supervised by Dr. Sara Kališnik.

Selected Programs

August 2026 **Illiad Intensive, Berkeley, CA**
Awarded a spot on the selective Illiad intensive course to broaden my understanding of the technical AI alignment field. $5000 stipend.

Research Interests

Broadly speaking, I am interested in applying what are generally considered to be pure mathematical tools to machine learning and biomedicine. For example, much of my past work has been in topological data analysis, where I worked on developing new loss functions based on topological invariants in the hopes of creating ML models that optimize for certain geometric structures.

Currently, I primarily focus on applying mathematical tools to mechanistic interpretability. For instance I'm interested in studying the geometry of model internal representations.

Publications

†= first or joint first authorship.

- {Luis Scoccola, Siddharth Setlur}†, David Loiseaux, Mathieu Carrière, and Steve Oudot. **Differentiability and Convergence of Filtration Learning with Multiparameter Persistence.** International Conference on Machine Learning (ICML) 2024.

Work Experience

July 2026 – October 2026	**Research Fellow, BlueLight AI** Worked on understanding how geometry arises in internal model representations of LLMs spanning both theoretical and empirical work. We aim to understand the various ways geoemtry manifests itself (e.g. due to co-occurence statistics in data, as a means of performing a task like linebreaking).
Oct 2025 – March 2026 .	**Research Assistant, Quantum Software Lab, University of Edinburgh** Contributed mathematical expertise to the QSL project entitled *Quantum Machine Learning for Genomics: Assessing Real-World Potential and Constraints*.
Jan 2024 – April 2024	**Teaching Assistant, Department of Mathematics, University of Edinburgh** Tutored a weekly course on titled *Advanced Research Computing*, which aimed to introduce machine learning methods to pure mathematicians. In particular, I helped design the assignments and exercises for the topological deep learning section.
Sept 2023 – Dec 2023	**Teaching Assistant, Department of Mathematics, ETH Zürich** Teaching a twice-weekly exercise class for first-year Math and Physics students taking Dr. Alessio Figalli's Single Variable Analysis course.
Oct 2022 – Feb 2023	**Research Intern, Inria Saclay (France)** Supervised by Dr. Steve Oudot. Research in theoretical foundations of topological data analysis, namely the differentiability of new invariants for multiparameter persistence. Funded by National Institute for Research in Digital Science and Technology
June 2021 – Sep 2022	**Library Assistant, Mathematics Library, ETH Zürich** Responsibilities included helping select new books for the collection, opening and closing the library once a week, processing book returns, answering customer queries, and conducting inventory.

Workshops and Conferences Attended

Julu 2026	**Human-aligned AI Summer School**, Center for Theoretical Study, Charles University, Prague, Czech Republic.
Jan 2026	**Foundations of causal inference**, Isaac Newton Institute, Cambridge University, UK.
July 2025	**SIAM Conference on Applied Algebraic Geometry (AG25)**, University of Wisconsin-Madison, U.S.
November 2024	**Twentieth Meeting of the Applied Algebra and Geometry Research Network**, Oxford University

August 2024	**Spires 2024**, Oxford University Poster Session for *Differentiability and Convergence of Filtration Learning with Multiparameter Persistence* (joint work with Luis Scoccola, David Loiseaux, Mathieu Carrière, and Steve Oudot).
July 2024	**The Forty-first International Conference on Machine Learning**, Vienna Poster Session for *Differentiability and Convergence of Filtration Learning with Multiparameter Persistence* (joint work with Luis Scoccola, David Loiseaux, Mathieu Carrière, and Steve Oudot).
August 2023	**BIREP Summer School on persistence modules—The Interplay of Representation Theory and Topological Data Analysis** Contributed Talk: *Signed Barcodes for Multiparameter Persistence.*

Travel Funding Awarded

August 2025	**The Geometric Realization of AATRN (Applied Algebraic Topology Research Network)**, Chicago, U.S. Awarded USD 1000 towards travel and lodging. Declined due to time constraints.
August 2024	**Spires 2024**, Oxford University Awareded a travel fund of GBP 200 and accommodation covered.
August 2023	**BIREP Summer School on persistence modules—The Interplay of Representation Theory and Topological Data Analysis** Travel expenses, accommodation, and food covered.

Service & Volunteering

Feb 2024 –	**Organizer, AGATE Seminar** Jointly organized a weekly seminar focused on applications of algebra, geometry, and topology to biology.
October 2024 – Dec 2024	**Organizer, BioTop Reading Group** Organized a hybrid seminar at the University of Edinburgh focusing on applications of topology, geometry, and algebra to biology.
Feb. 2022 – July 2022	**Organizer, Zurich Undergraduate Colloquium in Math and Physics** Organized a weekly colloquium where undergraduates were given the opportunity to present talks about their semester/bachelor/master theses.